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Editorial

Navigating the switch from IVIG to FcRn inhibition in CIDP: Clinical insights

Chronic Inflammatory Demyelinating Polyneuropathy (CIDP) is a syndrome that comprises a heterogeneous group of rare, acquired, autoimmune, demyelinating polyneuropathies [1]. The pathophysiology of CIDP, while not very well elucidated, involves both humoral (antibodies, complement) and cellular (autoreactive T cells) immune factors, with macrophage-induced demyelination [2]. CIDP is a treatable condition, and patients can have full resolution of symptoms, especially when treated early and adequately. Unfortunately, the diagnosis can sometimes be difficult, especially in CIDP variants. We often see patients with delayed diagnosis or inadequate therapy resulting in irreversible axonal loss and residual disability.

For decades, steroids and IVIG have been the main treatments for CIDP. Plasma exchange is also sometimes offered. For "refractory" CIDP, rituximab and cyclophosphamide can be used with variable degrees of success [3]. Steroid and IVIg are broad immunomodulatory agents with significant effects on multiple aspects of the immune system. To name just some of the documented effects of IVIg: blocking FcγR expression on monocytes or macrophages, regulating autoreactive B and T cells, modulating production of cytokines and cytokine antagonists, biding to complement and preventing membrane attack complex [4]. Most patients with CIDP will experience functional benefit with first line treatment. In fact, lack of response should trigger reconsideration of a CIDP diagnosis. However, there are significant downsides to these use of steroids and IVIG. Long term side effects of steroids are many. IVIG, while generally better tolerated also has potential adverse effects ranging from mildly bothersome to dangerous. Headache and nausea are common but can be mitigated by pretreatment (acetaminophen, steroids, and/or diphenhydramine), a small percentage of people can develop deep venous thrombosis (also mitigated by reducing the infusion rate), and some patients may have difficult venous access (mitigated by subcutaneous administration of IG or insertion of a port) [5]. Perhaps, the greatest downside of IVIG, is the infusion time (several hours per session every 3–4 weeks).

Fortunately, we now have a new approved treatment option for CIDP: subcutaneous Efgartigimod, a neonatal Fc receptor (FcRn) antagonist. FcRN antagonists bind to Fc region of the IgG molecule on the neonatal FcRn, preventing IgG recycling and increase its degradation. After the success of FcRn antagonists in the treatment of myasthenia gravis (MG), there has been significant interest in use of these therapies in CIDP [6,7]. However, in contrast with MG, where most patients have IgG antibodies against acetylcholine receptors, different immunopathological mechanisms contribute to CIDP pathogenesis. The use of FcRN inhibition of IgG in CIDP is a bet on a narrow immunomodulatory approach to a disease with a complex underlying immune pathophysiology. A winning bet in the use of FcRN in CIDP was placed in the ADHERE trial of subcutaneous efgartigimod. It is notable however, that the trial had 2 stages, with the first stage being an open label stage in which all participants who received efgartigimod. Importantly, only participants who improved on efgartigimod were transitioned to the 2nd stage, which was a randomized-withdrawal, double-blind, placebo-controlled trial stage that showed a higher relapse rate for participants receiving placebo versus those receiving subcutaneous efgartigimod (hazard ratio 0·39 [95 % CI 0·25–0·61]; $p < 0{\cdot}0001$.

While the trial was successful, there is some clear indication from the open label portion that not all patients with CIDP respond to FcRn therapy. Consequently, the case series reported by authors Levine and Muley: "Early deterioration of CIDP following transition from IVIG to FcRn inhibitor treatment", should not come as a total surprise [8]. Levine and Muley report their experience with 9 CIDP patients who were stable on IVIG and transitioned to subcutaneous Efgartigimod. Five of the 9 patients neither improved nor declined with the transition to subcutaneous Efgartigimod, suggesting that in those 5 patients, the main driver of the disease may have well been IgG autoantibodies. However, the other 4 patients experienced rapid and significant deterioration following transition from IVIG to subcutaneous Efgartigimod. Two patients sustained leg fractures following falls. Most of the patients required hospitalization and rescue treatment with plasma exchange, IVIG and prednisone. None of the patients have returned to their prior baseline, yet. Three patients transitioned from IVIG to subcutaneous Efgartigimod because they had residual weakness and hoped for additional improvement and 1 patient had fatigue as side effect from IVIG. The time to switching ranged from 12 days to 6 weeks after last IVIG dose. Worsening of symptoms occurred between 2 and 9 days after first efgartigimod administration. Patients were on IVIG for an average of 5.5 years and they had mild disability while on IVIG (INCAT 1–2) [8].

Levine and Muley's report suggests that some patients with CIDP who are stable on IVIG may transition successfully to subcutaneous Efgartigimod, while others may not. This aligns with our experience as well. Argenx, the maker of subcutaneous Efgartigimod, is currently planning a "Study to Assess Adults with CIDP Transitioning from IVIg to subcutaneous Efgartigimod" which plans on enrolling 25 patients and "will measure how adults with CIDP receiving IVIg treatment adjust to efgartigimod". (ClinicalTrials.gov ID NCT06637072) [9] In the meantime, post hoc analysis of the original ADHERE study, on those patients who transitioned from IVIG to subcutaneous Efgartigimod will hopefully be done and shared.

Finally, we note that 3 of 4 patients in the report by Levine and Muley, switched from IVIG to subcutaneous Efgartigimod in the hope of further improvement. No head-to-head studies compared IVIG and efgartigimod in CIDP patients and it is not clear from the ADHERE study

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that patients who switched from IVIG to subcutaneous Efgartigimod improved further. Currently, the main clear benefit of subcutaneous Efgartigimod over IVIG (or subcutaneous IG) is the rapid administration of therapy (around 90 s) and a favorable side effect profile (e.g., fewer headaches). Prior to switching therapy in patients with CIDP who are stable on IVIG to subcutaneous Efgartigimod or other therapies (e.g., steroids or subcutaneous IG), weighing the pros and cons with patients is essential, as there is always a risk of deterioration. Generally, there should be a compelling reason to switch therapies when patients are doing well on IVIG, particularly if the alternative therapy is not clearly superior.

CRediT authorship contribution statement

Chafic Karam: Writing – original draft.

Declaration of competing interest

Dr. Karam has served on the CIDP Diagnostic Adjudication Committee in the ADHERE Trial, has served as consultant for Argenx, and received research funding from Argenx.

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